

Module 8

Assignment 1 : Written

Q.1 What is CSS? Why is CSS required in web development?

→ CSS (Cascading Style Sheet) is a style sheet language used to control the appearance of HTML elements such as colors, fonts, layout, spacing

CSS is Required for

- 1> It separates content from design
- 2> makes websites attractive and user-friendly
- 3> Helps maintain consistency across multiple pages
- 4> Reduce code repetition and improves performance

Q.2 Differences between HTML & CSS

HTML

- Defines structure of web page
- Uses tags like `<p>`, `<h1>`
- Cannot control design fully
- mandatory for web page

CSS

- Define style of web page
- Uses selectors like `p`, `h1`
- Controls colors, layout, fonts
- Optional but important

Q.3 Basic syntax of CSS with example

Syntax :-

```
selector {  
    property: values;  
}
```

eg. P {

color: blue;

font-size: 16px;

}

Q4 Types of CSS

1> Inline CSS

Applied directly inside HTML tag.

* Advantages:

- Quick styling
- Highest priority

* Disadvantages

- Difficult to manage
- No reusability

2> Internal CSS

Written inside <style> tag in <head>

* Advantages

- Suitable for single page
- Easy to modify

* Disadvantages

- Not reusable for multiple pages.

3> External CSS

Written in a separate .css file.

* Advantages

- Reusable
- Easy maintenance
- Professional approach

* Disadvantage

- Extra file required.

Q.5 CSS comments & importance

- CSS Comments are written using:

`/* This is a comment */`

- Importance :

- 1> Improves readability.
- 2> Helps in debugging
- 3> Explain code logic

Q.6 Explain the terms

- CSS Selector : Selects HTML element (eg. p)
- CSS Property : Style Attribute (eg. color)
- CSS value : Assigned value (eg. red)

Q.7 Priority order CSS

- Inline CSS (Highest)
- Internal CSS
- External CSS
- Browser default (Lowest)

Q.8 Why External CSS is preferred?

- Same CSS used for multiple pages.
- Easy maintenance
- Faster page loading
- Cleaner HTML code.

- Q9. Problems with only inline CSS
- Default to maintain large website.
 - Code repetition.
 - Poor readability.
 - No reusability.

Q.10 Real-life use of CSS

- Design layouts
- Create responsive websites
- Apply animations
- improve user experience.

Assignment 1 : Written.

Q.1 What is CSS Selector? Why are Selector important?
A CSS Selector is used to select HTML elements that we want to style.

- Applies style to specific elements
- Controls website appearance.
- Makes CSS powerfull and flexible.

Q.2 Element selector with syntax and example
Element selector selects HTML elements by tag name.

Syntax:-

```
element {
  property: value;
}
```

example:

```
p {
  color: red;
}
```

Q.3 Class Selector and Id selector

- Class Selector :- Styles multiple elements.
Syntax:- • Class Name
- Id Selector :- Styles one required element.
Syntax:- #idName.

Class Selector

Id Selector

- Can be used multiple times - Used only once
- (.) - (#)

Q.4

Grouping Selector:-

A grouping Selector applies Same style to multiple element.

Syntax:-

h1, h2, p {

color: blue;

}

Used :- Reduce code repetition

Q.5

~~Reduces code repetition~~

Nested (descendant) selector.

→

Select elements inside another element.

ex:-

div p {

color: green;

}

Q.6

Attribute Selector:-

Attribute selectors style elements based on attributes.

Types:-

1) [attr] → [disabled]

2) [attr = value] → [type = "text"]

3) [attr^ = value] → [href^ = "https"]

Q.7

Pseudo - classes

Pseudo - classes define special states of elements

ex: :hover, :focus, :active, :visited

Q.8 Pseudo-Element

Pseudo - elements style specific parts of elements.

ex:- :: first-letter
 :: first-line
 :: before
 :: after

Q.9 Difference between pseudo-class and pseudo-element

Pseudo Class

- Using single colon
- Styles state
- Ex :: hover

Pseudo element

- Use double colon
- Style part of element
- Example :: first-letter

Q.10 Real life use of CSS selector

- Style buttons on hover
- Design forms
- Highlight menu items
- Create responsive layouts

Module 10

Assignment 1: Written

Q.1 Difference between HEX & RGB

① HEX :-

Uses hexadecimal values starting with #

ex: HEX : #FF0000 (Red)

② RGB :-

Uses Red, Green, Blue values (0-255)

ex: -

RGB: rgb(255, 0, 0)

Q.2 RGBA and Alpha Value

RGBA includes alpha ~~value~~ (transparency)

Alpha Value Controls opacity

Range: 0 (Full transparent) to 1 (Fully opaque)

eg. rgba(0, 0, 0, 0.5)

Q.3 HEX code for white and Black

- White: #FFFFFF
- Black: #000000

Q.4 % Unit For width

Percentage width is calculated based on parent element's width.

ex:- width: 50%.

Q.5 Difference between em & rem

- em → Relative to parent font-size
- rem → relative to root (html)

Q.6 When to prefer rem

- For Consistent, Scalable UI
- When you want predictable sizing.
- In responsive design.

Q.7 vh and vw

- VH → Viewport height
- Vh → Viewport width
- 100vh → Full Screen section
- 50vh → responsive card.

Q.8 Cover vs Contain.

Cover

- Fill entire container
- May crop image

Contain

- Fit whole image.
- No cropping

Q.9 background-repeat types

- repeat → repeat both sides
- no-repeat → show once
- repeat-x → horizontal.
- repeat-y → vertical.

Q.10 Background ~~image~~ Shorthand

Shorthand combines multiple properties

eg.

background: url(bg.jpg) no-repeat center/cover

Q.1 What are the text properties in CSS? Why are they important?

~~The~~ Text properties in CSS are used to control the appearance of text such as Color, alignment, spacing, decoration, and transformation.

Importance:-

- Improves readability
- Enhances user experience.
- Gives professional look to website.

Q.2 Explain Color property in CSS. Ways to define Color

The color property sets the text color.
Ways →

- 1) Color- Name - red
- 2) HEX - #ff0000
- 3) RGB - rgb(255, 0, 0)
- 4) RGBA - rgba(255, 0, 0, 0.5)

Q.3 Explain text-align
text-align aligns text horizontally.

Values & use

- left - default reading text
- right - side content
- center - headings
- justify - articles & blogs

Q.4 What is text-decoration?

- Used to add or remove decorative lines

Values

- none - remove underline
- underline - links
- line-through - old price
- overline - emphasis.

Q.5 Explain text-transform

Control text case.

Values.

- uppercase
- lowercase
- capitalize

Makes headings uniform and professional.

Q.6 What is line-height

- Controls space between lines,

- Improves readability, especially for long text.

Q.7 Font properties

a) Font-family :- Specifies font type.

b) Font-size :- Controls text size.

Q.8 What is font-weight?

Control text thickness

Keyword values: normal, bold, lighter

Numeric values - 100-900

Q.9 What are google fonts?

- Free web fonts provided by Google.

- Advantages -

Modern look, Easy to Use, Free

- DisAdvantages: -

Internet dependency, Slight performance impact.

Q.10 Text alignment and spacing.

- Proper alignment & spacing

Improve readability, Reduce eye strain, Should be used carefully to avoid clutter.

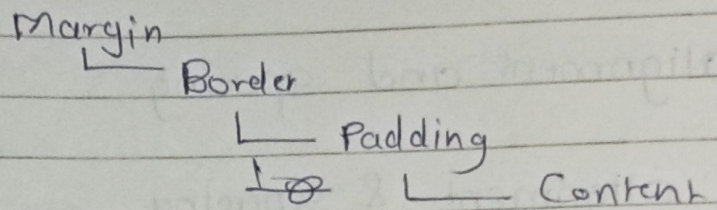
Module 12

Assignment 1: Written

Q.1 What is CSS box model?

The CSS Box model is a layout model that represents every HTML element as rectangular box.

- 1) Content - actual text or image.
- 2) Padding - Space around content
- 3) Border - Outline around padding
- 4) Margin - Space outside border.



Q.2 Content area in Box Model

The content area contains text, images, or other elements.

- Width → Controls horizontal size.
- height → Controls vertical size. Padding and border are added outside content.

Q.2 What is padding?

Padding is space between content and border.

- padding-top
- padding-right
- padding-bottom

- padding-left
- padding (shorthand)

Q.4 Border property

Border is line drawn around padding

Style with ex.

- solid → continuous line
- dashed → broken line
- dotted → dotted line

Border: 2px solid black;

Q.5 What is margin?

Margin space outside the border.

Diff

- 1) padding → inside border
- 2) margin → outside border.

Q.6 What is box sizing

Controls how width & height are calculated,

- Content-box: size excludes padding & border.
- Border-box: size includes padding & border

Q.7 Why border-box is preferred?

- Predictable layouts
- Easier calculations
- Better responsive design.

Q-8 Margin collapse
Margin collapse occurs when vertical margins of adjacent elements overlap.

Occurs between:

- Heading and paragraph
- Parent and first/last child

Q-9 Margin Shorthand
Used to apply margin in one line.

Ex: - `margin: 20px;`
`margin: 10px 20px;`
`margin: 10px 15px 20px 25px;`

Q-10 Real-life use of Box-model

- Cards & containers
- Website layouts
- Navigation menus
- Forms & buttons